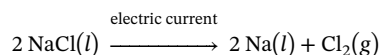
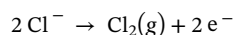


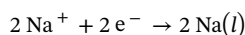
The overall reaction for the electrolysis of molten sodium chloride,



can be expressed as two separate, simultaneous redox processes (Reactions 1 and 2):



**Reaction 1**



**Reaction 2**

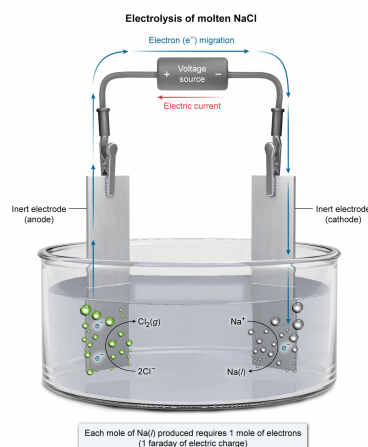
Given that the reduction of 2 moles of  $\text{NaCl}(l)$  requires 2 moles of electrons (a 1:1 ratio), what is the amount of electric charge required to produce 0.80 mol of  $\text{Na}(l)$  during the electrolysis?

- ✓ ☒ A. 0.8 faradays [55%]  
☐ B. 1.6 faradays [32%]  
☐ C. 2.4 faradays [9%]  
☐ D. 3.2 faradays [2%]

Correct

55% answered correctly

**Explanation:**



An **electrolytic cell** can be conceptualized as a **galvanic (voltaic) cell** that is operating in reverse. In an electrolytic cell, electric current from an external source (such as a battery) is supplied to drive a **nonspontaneous, endergonic** chemical reaction ( $\Delta G > 0$ ). The operation of an electrolytic cell causes **electrolysis**, which is the electrochemical **decomposition** of a compound.

During electrolysis, electron-rich species (such as anions) are **oxidized** at the electron-deficient anode. For example, in the electrolysis of molten  $\text{NaCl}$ ,  $\text{Cl}^-$  is oxidized to  $\text{Cl}_2$  gas (Reaction 1). Conversely, electron-deficient species (such as metal cations) are reduced at the electron-rich cathode. In the example of the electrolysis of molten  $\text{NaCl}$ ,  $\text{Na}^+$  is reduced to  $\text{Na}$  metal (Reaction 2). The net reaction for these simultaneous processes results in a decomposition reaction.

Based on the mole ratios obtained from the **stoichiometric coefficients** of the overall balanced reaction, the number of moles of  $\text{NaCl}(s)$  that must undergo electrolysis to form 0.80 mol  $\text{Na}(l)$  is given by:

$$0.80 \text{ mol Na}(l) \times \frac{2 \text{ mol NaCl}(l)}{2 \text{ mol Na}(l)} = 0.80 \text{ mol NaCl}(l)$$

Reactions 1 and 2 show that 2 moles of electrons are required for every 2 moles of  $\text{NaCl}(l)$  that undergo electrolysis (ie, the proportion reduces to a 1:1 ratio). Consequently, the electrolysis of 0.80 mol  $\text{NaCl}(l)$  requires the same number of moles of electrons:

$$0.80 \text{ mol NaCl}(l) \times \frac{2 \text{ mol e}^-}{2 \text{ mol NaCl}(l)} = 0.80 \text{ mol e}^-$$

The charge used during electrolysis can be measured in a unit called the faraday, where **1 faraday** equals the **electric charge** present in **1 mole of electrons**. Accordingly, the number of faradays present in 0.80 mol  $\text{e}^-$  is:

$$0.80 \text{ mol e}^- \times \frac{1 \text{ faraday}}{1 \text{ mol e}^-} = 0.80 \text{ faradays}$$

Therefore, 0.80 faradays of electric charge are required to produce 0.80 mol of  $\text{Na}(l)$  by electrolysis.

**(Choice B)** Using 1.6 faradays of electric charge would produce 1.6 mol  $\text{Na}(l)$ .

**(Choices C and D)** The electrons removed from the  $\text{Cl}^-$  anions in Reaction 1 are the same electrons used to reduce the  $\text{Na}^+$  cations in Reaction 2. Consequently, the ratio of moles of electrons to moles of  $\text{NaCl}$  is 1:1 rather than 3:1 or 4:1.

**Educational objective:**

In an electrolytic cell, electric current from an external source is supplied to provide the necessary energy to perform a nonspontaneous (endergonic), electrochemical decomposition (electrolysis) of a compound. The charge used during electrolysis can be measured in a unit called the faraday, where 1 faraday equals the electric charge present in 1 mole of electrons.

**Time spent:**0 sec

**Subject:**  
General Chemistry

**QID:**402713

**System:**  
Solutions and Electrochemistry